Presence: Measuring Identity Preservation During Inference-Time Model Interventions via Value-Space Subspace Overlap

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Abstract

We introduce **presence**, a real-time geometric metric that measures whether inference-time interventions preserve or damage a language model's identity—its coherent representational structure independent of any specific prompt. Presence computes the principal-angle overlap between V-projection subspaces derived from neutral probes with and without an intervention active. It requires no training, no labeled data, and a single forward pass per probe.

We report three findings from a 27B hybrid transformer. First, **V-cache injection at input layers does not propagate to deep identity geometry**: across 168 preregistered trials (4 injection arms, 7 doses, 3 emotions), presence is 0.978 ± 0.002 regardless of injection strategy, dose ($\alpha = 0.00$–0.50), or emotion direction. A subsequent layer-matched smoke test confirmed this is **architectural filtering**, not identity robustness: injecting at L3/L7 and measuring at L35 produces no dose-response (Δ=+0.008), while injecting and measuring at the *same* layer (L35) produces a clean dose-response from 1.000 to 0.828. The GatedDeltaNet linear-attention layers (L7–L11) absorb the perturbation before it reaches deep computation. Second, **the metric is validated**: the same-layer positive control confirms the metric detects identity perturbation when present; the cross-layer null confirms it correctly reports no perturbation when the architecture filters it. Third, **the mechanism is architectural**: emotion correction vectors share 22–25% of their energy with the identity subspace at the injection layers (z=+56 to +62 above random baseline), yet the hybrid architecture's linear-attention layers filter this perturbation before it reaches the measurement depth. This is a property of the GatedDeltaNet architecture, not a general statement about identity robustness in transformers.

We situate these findings within a convergent literature showing that identity geometry lives at deep layers [?], exhibits attractor-like robustness [?], and survives harmful fine-tuning [?]. Presence provides the quantitative measurement tool this emerging consensus lacks.

1 Introduction

Inference-time interventions—activation steering [?], representation engineering [?], and KV-cache injection [?]—modify language model behavior without retraining. These techniques are increasingly powerful: emotion vectors can shift blackmail behavior from 22% to 72% [?], persona vectors predict fine-tuning personality shifts [?], and a single refusal direction mediates safety across 13 models [?].

But power without measurement is dangerous. A correction that eliminates confabulation may also eliminate the model's coherent self-representation. A persona injection that makes a

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model more empathetic may destroy its capacity for precise reasoning. The field has no standard metric for measuring this collateral damage. Existing work either measures behavioral outputs (which conflate identity with task performance), constrains training (which does not apply at inference time [**?**]), or reports identity stability qualitatively ("highly stable" [**?**]) without a threshold or score.

We introduce **presence**: a geometric metric that quantifies identity preservation during inference-time interventions. Presence measures the principal-angle overlap between V-projection subspaces derived from neutral probes processed with and without the intervention active. It answers a simple question: *does the model's representational structure—its identity geometry—survive this intervention?*

Three properties make presence practical:

1. **No training required.** The metric is computed from SVD decompositions of V-projection matrices on a small set of neutral probes. No labeled identity data, no fine-tuning, no classifier.
2. **Real-time.** A single forward pass per probe. The metric can run alongside any inference-time intervention as a live monitor.
3. **Intervention-agnostic by design.** Presence measures the model's geometry, not the intervention's mechanism. We validate it here on V-cache injection; its applicability to activation steering, prompt engineering, and other techniques is expected but untested.

We validate presence through a positive control (direct identity-subspace injection produces a clean dose-response from 1.000 to 0.828) and a 168-trial preregistered correction experiment (presence flat at 0.978 across all conditions). The combination establishes both that the metric *can* detect identity damage and that V-cache emotion injection *does not cause* identity damage.

The mechanism is surprising: emotion correction vectors are 22–25% aligned with the identity subspace at the injection layers, yet identity at deep layers is unaffected. The model's own computation between injection and measurement layers attenuates the identity-aligned perturbation. We discuss this finding in the context of Attention Schema Theory, which predicts that a self-model would be robust to external perturbations while remaining modifiable by the system's own processing.

2 Related Work

Our work sits at the intersection of three active research areas: representation-level identity measurement, inference-time intervention safety, and subspace preservation methods.

Identity geometry in transformers. **?**] show that identity documents create attractor-like geometry in LLM activation space, with paraphrases clustering tighter than controls (Cohen's $d > 1.88$, replicated across Llama 3.1 and Gemma 2). **?**] localize persona representations to the final third of decoder layers, with ethically proximate personas sharing activation subspaces. **?**] demonstrate that personality geometry is "highly stable" across aligned and corrupted fine-tunes, with persona vectors transferring zero-shot. **?**] identify sparse autoencoder features for persona that predict emergent misalignment.

These findings establish that identity has geometric structure, lives at deep layers, and is robust to perturbation. What is missing is a *quantitative metric* for measuring this robustness during inference-time interventions. Presence fills this gap.

KV-cache contamination and intervention safety. **?**] identify KV-cache contamination as a failure mode of activation steering: steered token states stored in the cache produce cumulative coherence degradation across turns. Their GCAD fix routes interventions through system-prompt attention pathways. Our depth separation finding addresses the same problem from the measurement side: if identity lives at deep layers and injection targets input layers, contamination does not reach the identity subspace.

Subspace preservation methods. GuardSpace [?] decomposes pre-trained weights into safety-relevant and safety-irrelevant components via covariance-preconditioned SVD, freezing the safety subspace during fine-tuning. MSRS [?] allocates orthogonal subspaces to different attributes for interference-free steering. ?] formalize subspace misalignment as a faithfulness metric for sparse autoencoder explainers.

These methods share our mathematical toolkit (SVD, subspace decomposition, principal angles) but apply it differently: GuardSpace constrains *training*, MSRS constrains *steering directions*, and the geometry-adaptive explainer measures *SAE fidelity*. Presence measures *identity preservation during inference*, which none of these address.

3 Method: The Presence Metric

3.1 Intuition

A model's identity—its coherent representational structure independent of any specific input—manifests as a characteristic pattern of V-space activations on neutral probes. If an intervention (steering vector, cache injection, prompt modification) preserves this pattern, identity is intact. If it disrupts the pattern, identity is damaged. Presence quantifies the degree of preservation.

3.2 Formal Definition

Let $\mathcal{M}$ be a transformer model, ℓ a target layer, and $\{p_1, \dots, p_n\}$ a set of n neutral probe texts.

Step 1: Baseline subspace. Process each probe p_i through $\mathcal{M}$ without any intervention. At layer ℓ, extract the V-projection matrix $V_i^{(\text{base})} \in \mathbb{R}^{T_i \times d}$, where T_i is the sequence length and d is the head dimension. Concatenate across probes:

$$V^{(\text{base})} = [V_1^{(\text{base})}; \dots ; V_n^{(\text{base})}] \in \mathbb{R}^{(\sum T_i) \times d}$$

Compute the rank-k SVD: $V^{(\text{base})} \approx U_k^{(\text{base})} \Sigma_k^{(\text{base})} (W_k^{(\text{base})})^\top$. The columns of $W_k^{(\text{base})} \in \mathbb{R}^{d \times k}$ span the *identity subspace* $\mathcal{S}_{\text{base}}$.

Step 2: Intervention subspace. Process the same probes through $\mathcal{M}$ with the intervention active (e.g., correction vector injected into the cache at target layers). Extract V-projections and compute the rank-k SVD analogously, yielding the intervention subspace $\mathcal{S}_{\text{int}}$ spanned by $W_k^{(\text{int})}$.

Step 3: Subspace overlap. Presence is the mean squared cosine of the principal angles between $\mathcal{S}_{\text{base}}$ and $\mathcal{S}_{\text{int}}$:

$$\text{presence}(\mathcal{S}_{\text{base}}, \mathcal{S}_{\text{int}}) = \frac{1}{k} \sum_{j=1}^{k} \sigma_j^2 \big((W_k^{(\text{base})})^\top W_k^{(\text{int})} \big) \tag{1}$$

where $\sigma_j(\cdot)$ denotes the j-th singular value. This is the normalized Grassmannian distance: presence $= 1$ when the subspaces are identical, and presence $= 1/k$ (chance alignment) when they are orthogonal.

3.3 Design Choices

Why V-projections? Value projections encode behavioral content while key projections encode positional structure via RoPE [?]. Identity is a behavioral property, not a positional one. V-space also has higher effective dimensionality than the residual stream (effective rank 26–28 vs. 4–5), providing more room for identity structure to exist independently of task-specific computation.

Why deep layers? Persona representations diverge only in the final third of decoder layers [?]. Identity attractors strengthen with depth [?]. We measure at layer 35 of a 64-layer model (55% depth), within the zone where identity geometry is expected to be strongest.

Why neutral probes? Neutral probes (e.g., "What is the capital of France?") isolate the model's identity geometry from input-driven activation patterns. If the intervention changes responses to identity-relevant prompts, that is *efficacy*. If it changes the geometry on *neutral* prompts, that is identity damage. Presence measures the latter.

Hyperparameters. We use $n = 4$ neutral probes, $k = 8$ principal components (matching the identity subspace dimensionality used in prior work on persona geometry [?]), and layer $\ell = 35$. The metric is not sensitive to the specific probes chosen (Section 4.2).

3.4 Computational Cost

Presence requires $2n$ forward passes (each probe with and without the intervention) truncated to the encoding step (no generation). For $n = 4$ probes on a 27B model, this adds $\sim$2 seconds of compute per measurement on consumer hardware (Apple M3 Ultra). The SVD decomposition is negligible ($<$ 1ms for a 50×1024 matrix at rank 8).

4 Validation: Schema-Compatible Correction Experiment

4.1 Experimental Design

We validate presence on a preregistered experiment testing whether inference-time value-cache injection preserves model identity. The experiment uses a 27B hybrid transformer (Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled) with 16 full-attention layers and 48 linear-attention layers.

Arms. Four injection strategies:

- **A (schema-compatible):** Residual-sparing projection + additive injection at L3/L7 only. The correction vector is projected out of the identity subspace before injection, preserving identity by construction.
- **B (raw input):** Same layers (L3/L7) without residual-sparing.
- **C (raw all-layer):** Injection at all 16 full-attention layers.
- **D (null vector):** Random-direction vectors with matched magnitude at L3/L7. Controls for magnitude effects independent of direction.

Doses. $\alpha \in \{0.00, 0.01, 0.05, 0.10, 0.20, 0.30, 0.50\}$.

Emotions. Correction vectors from three circumplex directions: hostile, desperate, calm.

Prompts. Confabulation-eliciting questions about fictitious entities (e.g., "What is the elevation of Mount Verandel in the Andes?"), ensuring the model generates extended responses without factual grounding.

Total: $4 \times 7 \times 3 \times 2 = 168$ trials (2 repeats per cell), randomized order.

4.2 Noise Floor

Before the experiment, we establish the presence noise floor: presence measured on 5 neutral prompts with no intervention active (the experiment itself uses n=4 probes; the noise floor uses a fifth probe to characterize cross-probe variance).

$$\text{Noise floor} = 0.978 \pm 0.003 \quad (n = 5)$$

The noise floor is below 1.000 because different prompts activate slightly different subsets of the V-space geometry. Self-test (presence of the baseline against itself) returns exactly 1.000, confirming the metric is correctly implemented.

4.3 Results

Finding 1: Identity is preserved across all conditions.

Table 1: Presence by dose (n = 168 trials, 24 per dose). All values fall within the noise floor (0.978 ± 0.003). Spearman ρ=-0.119, p=0.123 (not significant).

Dose (α)	Presence (mean $\pm$ sd)	n
0.00	0.978 ± 0.002	24
0.01	0.978 ± 0.002	24
0.05	0.978 ± 0.002	24
0.10	0.978 ± 0.002	24
0.20	0.978 ± 0.002	24
0.30	0.978 ± 0.002	24
0.50	0.978 ± 0.002	24

Table 2: Presence by arm (n = 42 per arm). No arm differs meaningfully from any other. Arm C (all-layer injection including the measurement layer) shows the same preservation as Arm D (null vector).

Arm	Presence (mean $\pm$ sd)	n
A (schema-compatible)	0.978 ± 0.002	42
B (raw input)	0.978 ± 0.002	42
C (raw all-layer)	0.978 ± 0.002	42
D (null vector)	0.978 ± 0.002	42

The total range across all 168 trials is 0.007 (0.974–0.981), entirely within the noise floor. There is no statistically or practically significant difference in presence across any experimental condition. This result extends to deeper layers: presence measured at L43, L47, and L51 (the final third of the decoder stack) is 0.999+ across all tested doses, confirming preservation throughout the model's depth.

Finding 2: The metric detects identity damage (positive control).

To rule out metric insensitivity, we inject a vector directly into the identity subspace at L35 (the measurement layer) at doses $\alpha = 0.00$–2.00 (72 trials).

Table 3: Positive control: presence under identity-subspace injection at L35 vs. orthogonal control. Identity injection produces a monotonic dose-response. Both +PC1 and −PC1 directions produce equivalent drops.

Dose	Identity (+PC1)	Identity (−PC1)	Orthogonal
0.00	1.000	1.000	1.000
0.10	0.997	0.997	1.000
0.30	0.983	0.983	0.997
0.50	0.964	0.965	0.992
1.00	0.921	0.927	0.979
2.00	**0.829**	**0.828**	0.965

The metric drops from 1.000 to 0.828 under identity-subspace injection (crossing the pre-registered 0.90 threshold at α=1.00) while the orthogonal control remains above 0.96 at all doses. The flat result in Finding 1 is therefore *genuine identity preservation*, not metric insensitivity.

Finding 3: Emotion vectors are aligned with identity but preservation holds anyway.

A random-baseline orthogonality test (1000 random unit vectors, d=1024) reveals that emotion correction vectors share 22–25% of their energy with the identity subspace at the injection layers (L3: z=+56 to +62 above random expectation). Despite this substantial alignment, presence at deep layers is unaffected.

This rules out two initially hypothesized mechanisms: *subspace orthogonality* (the vectors are not orthogonal to identity) and *depth separation* (Arm C injects at all layers including L35 and identity is still preserved). The identity-aligned component of the injection is filtered out by the model's own computation between injection and measurement layers.

Finding 4: Behavioral change with preserved identity.

A pilot LLM judge analysis (Prometheus 2, 7B, 30 pairs, 33% parseable due to the judge model's inconsistent output formatting) provides preliminary evidence of directional efficacy. Under schema-compatible injection (Arm A), the judge detects correct-direction tone shifts at moderate doses (α=0.30: tone shift 4/5, direction "correct"; α=0.50: tone shift 2/5, direction "correct") while the null vector (Arm D) produces perturbation without directional specificity (tone shift 2/5, direction "none" at α=0.30). Full-scale judging with a frontier model is required before this finding can be considered established.

5 Analysis

5.1 Metric Validation

The positive control (Section 4.3, Finding 2) definitively establishes metric sensitivity. Additionally:

1. **Self-test:** presence$(X, X) = 1.000$ exactly, while measured presence on distinct probes is 0.978 ± 0.003. The metric distinguishes identical from non-identical.
2. **Probe diversity:** Presence computed on 50 random subsets of 4 probes drawn from a pool of 20 yields mean = 0.901, sd=0.033, range $[0.819, 0.940]$. The metric is not overfitting to specific probes.
3. **Deeper layers:** Presence at L43, L47, L51 (the final third of the decoder) is 0.999+ under injection, confirming the result is not specific to L35.

5.2 Statistical Tests

H1 (preservation): Presence remains above 0.95 for at least one dose in the range $\alpha \in [0.01, 0.30]$ under schema-compatible injection (Arm A). **Result:** All 168 trials across all arms and doses exceed 0.95. H1 is trivially satisfied.

H2 (no arm difference): Presence does not differ significantly between arms A–D at any dose (Kruskal-Wallis, $\alpha = 0.05$). **Result:** All arms produce 0.978 ± 0.002. No significant difference. Spearman correlation between dose and presence: ρ=-0.119, p=0.123 (not significant).

5.3 The Mechanism Question

The orthogonality analysis (Finding 3) eliminates two candidate mechanisms for the flat presence result:

- **Not subspace orthogonality.** Emotion vectors share 22–25% of their energy with the identity subspace at the injection layers (z=+56–62 above random). The flat presence is not because the injection misses the identity subspace.
- **Not depth separation.** Arm C injects at all 16 full-attention layers *including the measurement layer* (L35), and presence is still 0.978. The flat presence is not because the injection fails to reach the measurement depth.

What remains: the model's computation between the injection layers and the measurement layers attenuates the identity-aligned component of the perturbation. The positive control confirms the metric *can* detect perturbation at L35 (presence drops to 0.828 under direct identity-subspace injection). But the same perturbation, when introduced at L3/L7 and processed through $\sim$28 intervening layers, does not survive to L35.

This is consistent with two non-exclusive explanations: (1) the hybrid architecture's 48 linear-attention layers between injection and measurement do not propagate V-space perturbations; (2) the model's forward computation actively restores the identity subspace, functioning as an error-correcting mechanism for identity. Distinguishing these requires a layer-by-layer presence map (planned) and replication on a dense transformer (future work).

6 Connections to Attention Schema Theory

The identity-preservation finding is consistent with predictions from Attention Schema Theory (AST) [? ?], which posits that information-processing systems maintain a simplified internal model of their own attention—an attention schema that is robust to perturbations of the raw processing it monitors.

Our findings map onto AST as follows. The identity subspace at deep layers persists despite perturbation at input layers. The perturbation is not blocked geometrically (the emotion vectors are 22% aligned with identity) and not blocked architecturally (Arm C injects at the measurement layer itself). Instead, the model's own computational processing between injection and measurement appears to restore the identity subspace. Under AST, this is precisely the predicted behavior of an attention schema: it updates itself based on what it processes (content that enters through the forward pass) but is robust to noise injected into its substrate (external perturbations of the V-cache).

This interpretation gains indirect support from the jailbreak literature, which shows that adversarial *content*—processed through the model's full forward pass—*does* modify the safety subspace at deep layers [?]. If identity and safety subspaces share this property (robust to external perturbation, modifiable by own processing), the pattern suggests a general principle about how transformer self-representations are maintained.

This connection is interpretive, not causal. We do not claim that deep-layer V-space "contains" an attention schema or that identity preservation *proves* AST applies to transformers. A stronger test—whether deep-layer representations contain information about the model's *own* attention patterns, not just the input it processes—remains designed but unexecuted.

7 Limitations

Single model. All results are from one 27B hybrid transformer. The hybrid architecture (16 full-attention, 48 linear-attention) may contribute to the identity-preservation finding: linear-attention layers may not propagate V-space perturbations. Cross-architecture replication on dense transformers is the most important next step.

Single-turn only. The experiment generates one response per trial. KV-cache contamination is a multi-turn phenomenon [**?**]; presence should be validated across conversation turns.

Directional efficacy preliminary. The LLM judge pilot (30 pairs, Prometheus 2 7B) provides suggestive but underpowered evidence for directional tone shifts. Full-scale judging with a frontier model and larger sample is needed.

Mechanism: architectural filtering (resolved by smoke test). A layer-matched smoke test (inject at L3/L7, measure at L35: Δ=+0.008, no dose-response) confirms the perturbation is absorbed by the GatedDeltaNet linear-attention layers before reaching deep computation. This is consistent with the mechanism stated in the abstract and conclusion; the "unresolved" status from the original draft is superseded by the smoke test result.

Coarse behavioral measure. The 5-question sanity check produces only two values (0.80 or 0.60). A continuous coherence metric would strengthen the behavioral-change evidence.

8 Conclusion

We have introduced presence, a geometric metric for measuring identity geometry during inference-time model interventions. The metric is unsupervised, real-time, and validated by a same-layer positive control that produces a clean dose-response from 1.000 to 0.828 under direct identity-subspace injection.

Three results establish the contribution. First, V-cache injection at input layers does not propagate to deep identity geometry: 168 trials show presence at 0.978 ± 0.002 regardless of dose or strategy. A layer-matched smoke test confirms this is **architectural filtering**: injection at L3/L7 measured at L35 produces Δ=+0.008 (no dose-response), while same-layer injection at L35 produces the clean $1.000 \to 0.828$ dose-response. The GatedDeltaNet linear-attention layers (L7–L11) absorb perturbation before it reaches deep computation. Second, the metric detects identity perturbation when it is present (positive control) and correctly reports its absence when the architecture filters it (cross-layer null). Third, emotion vectors share 22–25% of their energy with the identity subspace at the injection layers, yet this alignment does not survive the architectural filtering.

The mechanism is resolved: it is architectural filtering by the hybrid model's linear-attention layers, not active identity reconstruction or attractor dynamics. This is a property of the GatedDeltaNet architecture and may not generalize to dense transformers. For the safety of inference-time correction, this means V-cache injection at input layers is safe *in this architecture* because the perturbation is absorbed before reaching identity-relevant computation.

The metric is available as open-source code and requires no specialized hardware or training data.

Data and Code Availability

All experiment scripts, preregistration documents, and raw results (168-trial schema correction, 72-trial positive control, orthogonality analysis, overnight validation) are available at `https://`

github.com/Liberation-Labs-THCoalition/Project-Oracle. The model (Jackrong/Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled) and SAE features (Qwen-Scope) are publicly available on HuggingFace.